THE ZERO PARADOX

ZP-D: State Layer (Hilbert Space)

Initial: April 2026 | Current: 1.16, October 2026

This document operates within functional analysis. It imports from ZP-A and ZP-B and constructs the Hilbert space state layer on top of them. No information theory from ZP-C is imported. Cross-framework synthesis is deferred to ZP-E.

Illustrated Companion: A paired ZP-D Illustrated Companion provides concrete examples and visual intuitions for the results here. Examples are kept separate from the formal layers to distinguish illustrative material from proofs.

I. Imported Structure

1.1 From ZP-A: Algebraic Structure of States

Import I-A — From ZP-A: Lattice Algebra
$(L, \vee, \bot)$: join-semilattice with bottom. Axioms A1–A4.
$\leq$ partial order: $x \leq y :\Longleftrightarrow x \vee y = y$ (D1, T1).
$\bot$ is the global minimum: $\bot \leq x$ for all $x \in L$ (T2).
State transitions are joins: $f(x) = x \vee \alpha$ (D2).
State sequences are monotone: $S_n \leq S_{n+1}$ (T3).
CC-1: $S_0 = \bot$ — Conditional Claim; modelling commitment.

1.2 From ZP-B: Topological Domain

Import I-B — From ZP-B: p-Adic Topology
AX-B1: Binary Existence Axiom.
MP-1: Minimality Principle.
T0: $p = 2$ derived from AX-B1 and MP-1.
Q_2 with 2-adic metric d (D1, D2).
T1: Ultrametric (strong triangle inequality).
T2: Every ball is clopen.
T3: Clopen gap at 0.

Import I-B — From ZP-B: p-Adic Topology

T5: Q_2 is totally disconnected.

C3: Snap is topologically irreversible (corollary of T5).

ε_0: Minimum viable deviation, universe-contingent parameter (D5).

II. The Hilbert Space State Layer

Definition D1 — State Layer H

$H = \mathbb{C}^n$ is a complex Hilbert space with orthonormal basis $\{e_0, e_1, \ldots\}$.

The foundational minimum is $n = 2$, corresponding to the two ontological states of the framework: $\bot$ (the bottom state, mapped to e_0) and ε_0 (the minimum state off $\bot$, mapped to e_1). These two orthogonal vectors express the binary existence/non-existence distinction that is the framework's central object. All further states are derived from this pair as joins in $(L, \vee, \bot)$ — they require no additional foundational dimension.

The framework's core claims (T4: snap produces orthogonal shift; T5: monotone norms) are established at $n = 2$. Extensions to higher n are consistent and natural: for level-k approximations using the clopen ball partition of Q_2 at depth k, $n = 2^k$. No foundational claim of the framework requires $n > 2$.

Remark R1 — Decoupling of Topological and State Layers

Q_2 and H are categorically distinct structures. Q_2 is a topological field; H is a Hilbert space over $\mathbb{C}$. They share no operations.

The transition operator T: $Q_2 \to H$ is the only bridge. It is constructed explicitly in T2.

No operation in H is assumed to inherit a topological property of Q_2 without proof. Every cross-layer claim must go through T.

III. The Transition Operator T: $Q_2 \to H$

3.1 The Design Commitment — Orthogonality

Design Principle DP-1 — Orthogonality as the Representation of Clopen Separation

Clopen separation in Q_2 (T3: 0 is clopen-separated from all nonzero elements; clopen balls are mutually separated) is represented in H by orthogonality: elements that are clopen-separated in Q_2 map to orthogonal vectors in H.

Motivation: Orthogonality in H is the natural algebraic analogue of topological separation. $\langle e_i, e_j \rangle = 0$ for $i \neq j$; two clopen balls are maximally distinct in the topological sense.

Status: DESIGN PRINCIPLE — DP-1 is chosen, not derived. It is the natural and consistent choice, stated explicitly. T4 and T5 below depend on DP-1 as a premise.

3.2 The Construction Target

Definition D2 — Transition Operator T (Requirements)

T: $Q_2 \to H$ must satisfy:

(i) $T(0) = e_0$ (0 of Q_2, read as $\perp$, maps to the designated base vector)

(ii) $T(\varepsilon_0) = e_1$ (minimum deviation maps to the first non-base vector)

(iii) T is injective on the clopen ball partition

(iv) If x and y are in disjoint clopen balls, then $\langle T(x), T(y) \rangle = 0$ (DP-1)

(v) $\|T(x)\| \geq \|T(0)\|$ for all x (norm-increasing: additive ontology preserved)

3.3 Existence of T

Theorem T2 — Existence of T (Basis Assignment)

There exists a function T: $Q_2 \to H$ satisfying all five requirements of D2.

Proof: Construct T by basis assignment. The clopen ball partition of Q_2 at level k consists of 2^k disjoint clopen balls. Assign each ball to a distinct basis vector of H. $T(x) = e_i$ where i is the index of the ball containing x.

(i) 0 maps to e_0 by assignment. ✓ (ii) ε_0 maps to e_1 by assignment. ✓ (iii) Distinct balls → distinct basis vectors. ✓ (iv) Disjoint balls → orthogonal basis vectors. ✓ (v) All basis vectors have norm $1 \geq \|e_0\| = 1$. ✓

Remark R3 — Topological Type of T

T is locally constant: it is constant on each clopen ball of Q_2. This follows directly from DP-1 — clopen separation maps to orthogonality, so T does not interpolate between basis vectors across ball boundaries.

Continuity: T is continuous from (Q_2, 2-adic topology) to H. The image of T is a discrete set of basis vectors $\{e_0, e_1, \ldots\}$. Each preimage $T^{-1}(e_i)$ is a clopen ball of Q_2, which is open in the 2-adic topology. Therefore preimages of open sets in H are open in Q_2, and T is continuous. ✓

Note on connected spaces: Q_2 is totally disconnected; $H = \mathbb{C}^n$ with the norm topology is path-connected. A continuous map from a totally disconnected space to a connected space need not be constant — it must only have a totally disconnected image. T's image is a discrete (hence totally disconnected) set of basis vectors, which is consistent with both the total disconnectedness of Q_2 and the path-connectedness of H as a whole.

3.4 Uniqueness of T

Proposition T3 — Uniqueness of T up to Unitary Equivalence

Any two operators T, T': $Q_2 \to H$ satisfying D2 are related by a unitary transformation U: H → H such that $T' = U \circ T$.

Proof: T and T' both assign e_0 to the image of 0, which is the unique additive identity (A4). The ball structure of Q_2 is fixed; only the labelling of basis vectors varies. A unitary map U taking T(0) to T'(0) and preserving orthogonality relations defines the equivalence. ✓

Remark R2 — What T Is Not

T is not a ring homomorphism. Q_2 has field operations; H does not. T does not preserve addition or multiplication from Q_2.

T is not a topological embedding. The topology of H is the norm topology; the topology of Q_2 is the 2-adic ultrametric. T is a structure-preserving assignment: ontological distinctions (clopen separation in Q_2) map to algebraic distinctions (orthogonality in H), as specified by DP-1.

Remark R-PA — What T Is, in Standard Terms (Prior Art)

Representing the clopen-ball structure of Q_2 by an orthonormal system in a Hilbert space is the standard p-adic construction, not a ZP novelty: the indicator functions of clopen balls form an orthonormal basis for locally constant functions (the van der Put basis), which refines to Kozyrev's p-adic wavelet basis of $L^2(\mathbb{Q}_2)$ — the analog of the Haar basis and an eigenbasis of the Vladimirov operator (Kozyrev, 2002). p-adic quantum mechanics builds Hilbert-space state models (the "p-adic qubit") on the same structure.

ZP-D's contribution is not the embedding itself but the specific reading in DP-1 — clopen separation ↦ orthogonality — that carries the Binary Snap into the state layer. T is an instance of the recognized construction, applied to the snap.

IV. The Binary Snap in H

Theorem T4 — Snap Produces Orthogonal Shift in H

The Binary Snap $0 \to \varepsilon_0$ in Q_2 maps to an orthogonal shift in H: $T(0) = e_0$ and $T(\varepsilon_0) = e_1$, and $\langle e_0, e_1 \rangle = 0$.

Proof: By D2(i), $T(0) = e_0$. By D2(ii), $T(\varepsilon_0) = e_1$. Since 0 and ε_0 are in disjoint clopen balls of Q_2 (T3), D2(iv) and DP-1 give $\langle T(0), T(\varepsilon_0) \rangle = \langle e_0, e_1 \rangle = 0$. ✓

Status: Derived — unconditional theorem given DP-1.

Proposition T5 — Monotone Sequences Map to Non-Decreasing Norms

Let (S_n) be a monotone state sequence in L (ZP-A T3). Then $\|T(S_n)\| \leq \|T(S_{n+1})\|$ for all n.

T5 norm result: Derived — unconditional given DP-1. Note: in the basis-assignment construction (D2), all T-images are unit basis vectors — all norms equal 1. The norm inequality $\|T(S_n)\| \leq \|T(S_{n+1})\|$ is therefore satisfied trivially ($1 \leq 1$). The Lean proof (t5_monotone_norms) verifies this tautology; it does not verify the ball-boundary argument. ✓

T5-b — Strict Orthogonality (non-trivial content, Lean-verified): For any strictly monotone state sequence — where $S_n \neq S_{n+1}$ — consecutive T-images are orthogonal: $\langle T(S_n), T(S_{n+1}) \rangle = 0$. Proof: $S_n \neq S_{n+1}$ implies they index distinct clopen balls. By DP-1, distinct ball-indices map to orthogonal basis vectors. Therefore $T(S_n) \perp T(S_{n+1})$. Lean: t5_strict_orthogonal (uses DP-1 via t2_orthogonal; axiom-free). ✓

T5-b is the load-bearing result: it captures the genuine content of the ball-boundary structure and is the Hilbert-space expression of ZP-B C3 (topological irreversibility) applied to consecutive states.

V. Open Items Register

Item	Status	Description
DP-1: Orthogonality commitment	Design Principle — explicit	Reclassified from Theorem T1. Orthogonality is chosen, not derived. Content unchanged.
T2: Existence of T	Closed	Basis assignment construction. All five requirements verified.
T3: Uniqueness of T	Closed	Unique up to unitary equivalence.
T4: Snap → orthogonal shift	Closed — unconditional	Proven from T2 and ZP-B T3. Depends on DP-1 as premise.
T5: Monotone norms + T5-b: Strict orthogonality	Closed — unconditional given DP-1	T5 norm result: trivially satisfied (all norms = 1 by construction). T5-b: distinct consecutive states produce orthogonal T-images — non-trivial, Lean-verified (t5_strict_orthogonal).

VI. Validation Status

Component	Status / Notes
$H = \mathbb{C}^n$ (D1)	Valid — Defined; foundational minimum n = 2 (binary existence/non-existence); extensions to $n = 2^k$ consistent; no core claim requires n > 2
Decoupling of Q_2 and H (R1)	Valid — Structural; Q_2 and H are categorically distinct; T is the bridge
Import I-A from ZP-A	Valid — Received; CC-1 reclassification noted
Import I-B from ZP-B	Valid — Received; MP-1 included; C3 noted
DP-1: Orthogonality	Valid — Design Principle; reclassified from T1; well-motivated and explicit
D2: T requirements	Valid — Defined; five requirements stated; all satisfied by T2
T2: Existence of T	Valid — Derived; basis assignment; all five requirements verified; R3 names topological type: locally constant, continuous
T3: Uniqueness of T	Valid — Proposition; derived; unique up to unitary equivalence
T4: Snap → orthogonal shift	Valid — Theorem; derived; unconditional; depends on DP-1
T5 / T5-b: Monotone norms + strict orthogonality	Valid — T5 norm result: tautology of construction (all norms = 1); Lean: t5_monotone_norms. T5-b: non-trivial Lean-verified result — distinct consecutive states produce orthogonal T-images via DP-1; Lean: t5_strict_orthogonal (axiom-free).